



Textile (TMS) Operational Insights

1. Fabric Inspection Pass Rate & Quality Deterioration

- **Critical Deficit:** Only **24.6%** of fabric rolls achieve a premium **"Grade A"** designation during inspection.
- **Substandard Bulk:** The vast majority of stock is classified as **Grade B (35.61%)** or **Grade C (39.78%)**. This reveals that **over 75%** of all fabric inputs entering production fail to meet standard specifications, introducing severe downstream defect risks across all active buyer accounts.

2. Dyeing Reprocess Rates

- **Benchmark Breach:** The overall dyeing reprocess rate stands at **12.3%**, which is more than double the maximum industry benchmark of **5.0%**.
- **Material Exceptions:** **Twill** fabric is identified as the weakest category at a **15.6%** reprocess rate, followed closely by **Rib (13.8%)** and **Single Jersey (11.8%)**. This dynamic effectively diminishes dyeing throughput by **over one in eight** batches, causing systemic waste of chemical inputs, water, and thermal energy.

3. Knitting Output & Mechanical Faults

- **Volume Constraints:** Knitting production experienced a sharp, multi-year decline in **2025** after peaking during the **2023–2024** timeframe. Overall knitting efficiency sits at **78.5%**, trailing the company target of **80%**. This drop highlights excessive machine downtime and stop minutes rather than permanent capacity degradation.
- **Defect Profiling:** Both **"Uneven Loop"** and **"Needle Mark"** faults surpassed **500** total occurrences. These are direct indicators of worn knitting needles and improper mechanical tension configurations, fueling the downstream inspection failures.

4. Fabric Shrinkage Levels

- **Escalating Costs:** Average shrinkage metrics have steadily risen from **6.2%** in **2022** to a stable ceiling of **6.4%–6.6%**, overshooting the acceptable **5.0%** ceiling. This excess shrinkage creates severe dimension inaccuracies in finished garments and forces artificial increases in fabric consumption to cover the variance, inflating raw material expenditure per unit.



Garments (GMS) Production Insights

5. The Cutting Room Bottleneck

- **Systemic Constriction:** Cutting achievement is severely bottlenecked at an average of **55.5%**. This metric is locked within an incredibly tight, invariant range of **55.47% to 56.61%** across all eight primary buyers.
- **The Downstream Cascade:** Because slightly more than half of the requested order volume is actually being cut, it creates an immediate starvation of inputs for the sewing lines, deflates total finishing volume, and acts as the primary root cause of delivery delays.

6. Sewing Efficiency vs. Floor Quality

- **The Trade-Off:** The sewing floor demonstrates an outstanding nominal efficiency rating of **99.3%**, indicating lines run at near-maximum capacity.
- **Defect Density:** This speed comes at a heavy cost to quality: the **Defects Per Hundred Units (DHU)** metric stands at **4.5**, severely breaching the target ceiling of **3.0**. Rapid sewing paired with weak inline inspection allows defects to slip downstream, artificially inflating the finishing rejection volume.

7. Output Attrition & Finishing Rejections

- **Widening Gaps:** Total produced quantities remain on a multi-year downward trajectory throughout **2025**. Realized output continuously fails to meet monthly targets, with the missing volume gap widening significantly in the latter months of the fiscal year.
- **Rejection Persistence:** Finishing reject rates show minor positive movement, falling from **3.54%** in **January** to **3.38%** by **December**. Despite a solid **96.5%** finishing achievement rate, the metric remained above the **3.0%** threshold all year, causing packing losses to accumulate over thousands of items monthly.

8. Commercial Shipment Risk

- **Late Deliveries:** On-time shipment performance is at a critically low **34.17%**, with only **205 out of 600** total shipments arriving on schedule.
- **Financial Exposure:** With **395** shipments arriving late due to the cutting room bottleneck, the plant faces severe commercial vulnerability—including order cancellations, financial penalties, and compromised long-term buyer retention.



Priority Focus Areas

The analytics indicate three immediate, non-negotiable operational priorities to safeguard business viability:

- **On-Time Shipments (34.17%):** Must be elevated immediately to mitigate commercial default risks and severe contract penalties.
- **Cutting Achievement (55.5%):** Must be expanded as the absolute primary physical bottleneck stifling the entire downstream sewing and packing floor.
- **Fabric Inspection Pass Rate (24.6%):** Upstream fabrication settings must be overhauled to stop substandard rolls (over 75% of total input) from passing into production.

Core Strategic Recommendations

-  **Implement Planned Needle & Tension Audits:** Establish strict preventative maintenance cycles on all knitting machines to keep "Uneven Loop" and "Needle Mark" defects below an acceptable threshold, lifting the Grade A fabric conversion rate.
-  **Re-engineer the Cutting Room Balance:** Conduct an immediate capacity-balancing study in the cutting section to clear the **55.5%** barrier, ensuring cutting output matches the **99.3%** absorption capability of the sewing floor.
-  **Enforce Inline Quality Controls in Sewing:** Restructure line incentive programs to reward quality alongside output speed, cooling the DHU metric down to the **3.0** target and protecting finishing yield.